LESSON 5.8

Determine Surface Area of Pyramids

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of a pyramid by adding the base area and the lateral faces.

WHY THIS MATTERS Your class is building a pyramid-shaped display to showcase the time capsule at the school entrance. The pyramid will be covered in gold leaf to make it shine. To figure out how much gold leaf you need, you must calculate the total surface area — the base plus all the triangular faces! The pyramid has a square base that is 10 inches on each side, and each triangular face has a slant height of 12 inches.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Pyramid *Spanish: Pirámide*	A solid with a flat bottom and triangle sides that meet at one point on top.	square base + 4 triangles
Slant height *Spanish: Altura inclinada (apotema)*	The height of a side triangle, measured along its slanted face.	used in ½ × b × slant
Lateral face *Spanish: Cara lateral*	A triangle side of a pyramid, not the bottom.	4 on a square pyramid
Base *Spanish: Base*	The flat bottom of a pyramid.	the square on the net
Apex *Spanish: Ápice*	The point at the top of a pyramid where the sides meet.	tip of the pyramid
Lateral area *Spanish: Área lateral*	The total area of just the side triangles, not the bottom.	For a square pyramid: lateral area = 4 × (½ × base edge × slant height)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Square base 10 cm; slant height 7 cm.

1. Base: 10 × 10 = 100 cm².
2. One triangle: ½ × 10 × 7 = 35 cm².
3. Four triangles: 140. Add: 100 + 140.

Answer: SA = 240 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller square pyramid: base 4 × 4, slant height 5. What is the base area?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base is a _____ with an area of _____ .
- Each triangular face has a base of _____ and a slant height of _____ .
- There are _____ triangular faces, so together they cover _____ .
- The surface area is _____ because I added _____ .

Word bank pyramid | base | slant height | lateral face | apex | triangle | square | congruent | add | surface area | square units | net

WATCH OUT!

A common mistake

A common mistake in Surface Area of Pyramids is finding the four triangular lateral faces correctly (½ × base edge × slant height, times 4) and then stopping

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the area of one lateral face for each pyramid.

Pyramid	Base Edge	Slant Height	Area of One Face
Pyramid 1	4 in	9 in	
Pyramid 2	6 cm	5 cm	
Pyramid 3	10 ft	8 ft	

STEPS TO TRY

1. Each row asks for the area of ONE triangular face, using its base edge and slant height.
2. Use ½ × base edge × slant height for each row — multiply the two numbers, then halve.
3. Pyramid 1: multiply 4 × 9 = 36, then take half. Write the result with in².

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

(A) 70 in²
(B) 95 in²
(C) 95 in³
(D) 120 in²

STEPS TO TRY

1. A square pyramid has TWO kinds of faces: one square base (5 × 5) and four identical triangles with slant height 7.
2. SA = base area + 4 triangle areas, where each triangle is ½ × base edge × slant height.
3. Base first: 5 × 5 = 25 in². Then one triangle: ½ × 5 × 7. Multiply that by 4 and add the base area.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

How many lateral (triangular) faces does a square pyramid have?

(A) 3
(B) 4
(C) 5
(D) 6

STEPS TO TRY

1. Lateral faces are the slanted triangles on the sides — the square base itself doesn't count.
2. Each side of the square base holds up exactly one triangle. Count the sides of a square.
3. Trace the base: it's a square. Count one triangle leaning against each side of that square.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the area of one triangular face with base 8 in and slant height 6 in?

(A) 14 in²
(B) 24 in²
(C) 24 in³
(D) 48 in²

STEPS TO TRY

1. You have one triangle: base 8 in, slant height 6 in. Triangle area is not just base × height.
2. Use area = ½ × base × height — multiply the two numbers first, then take half.
3. Multiply 8 × 6 = 48 first. Then take half of that result.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

Display B has the largest surface area (280 in²). What contributes more to the total — the base or the lateral faces? Why?

SENTENCE STARTER For Display B, the base area is ___ in² and the total lateral area is 4 × ___ = ___ in². The ___ contribute(s) more because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help